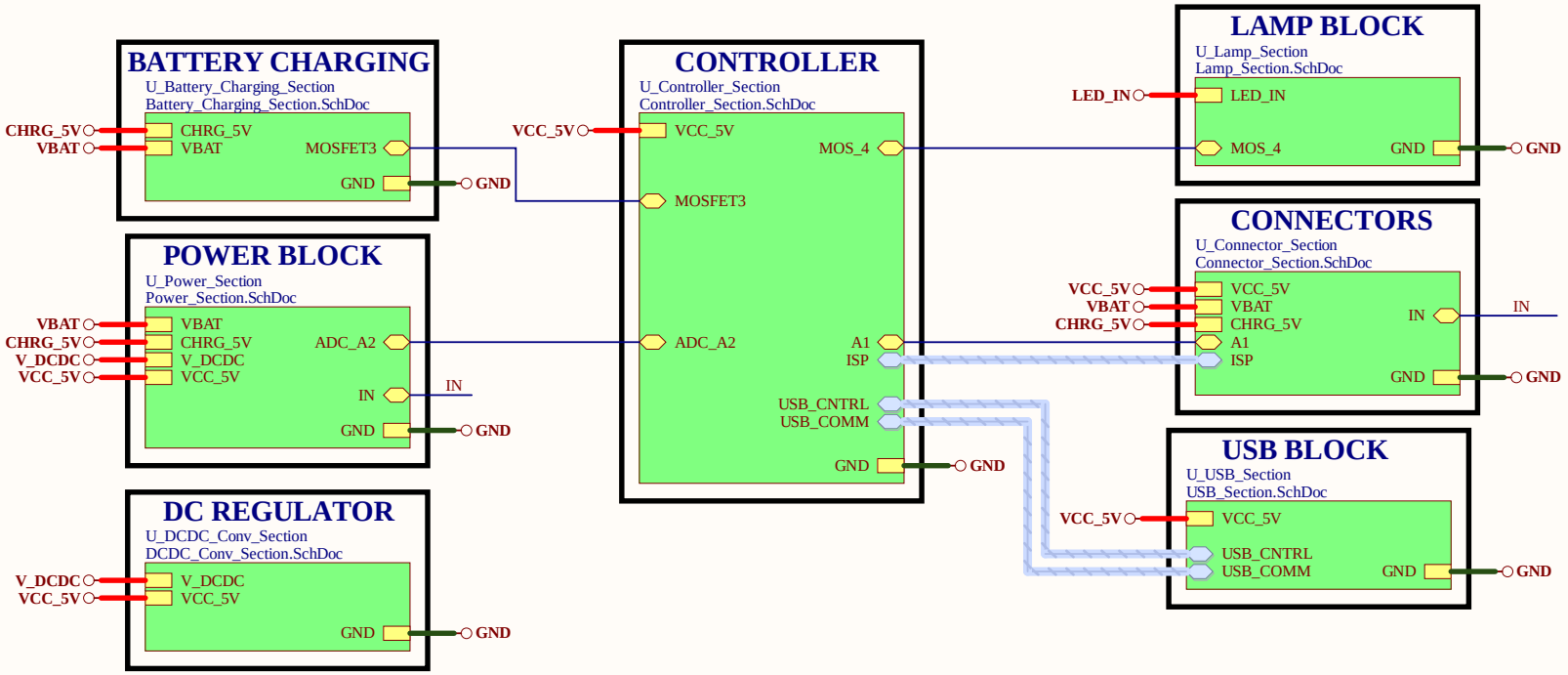
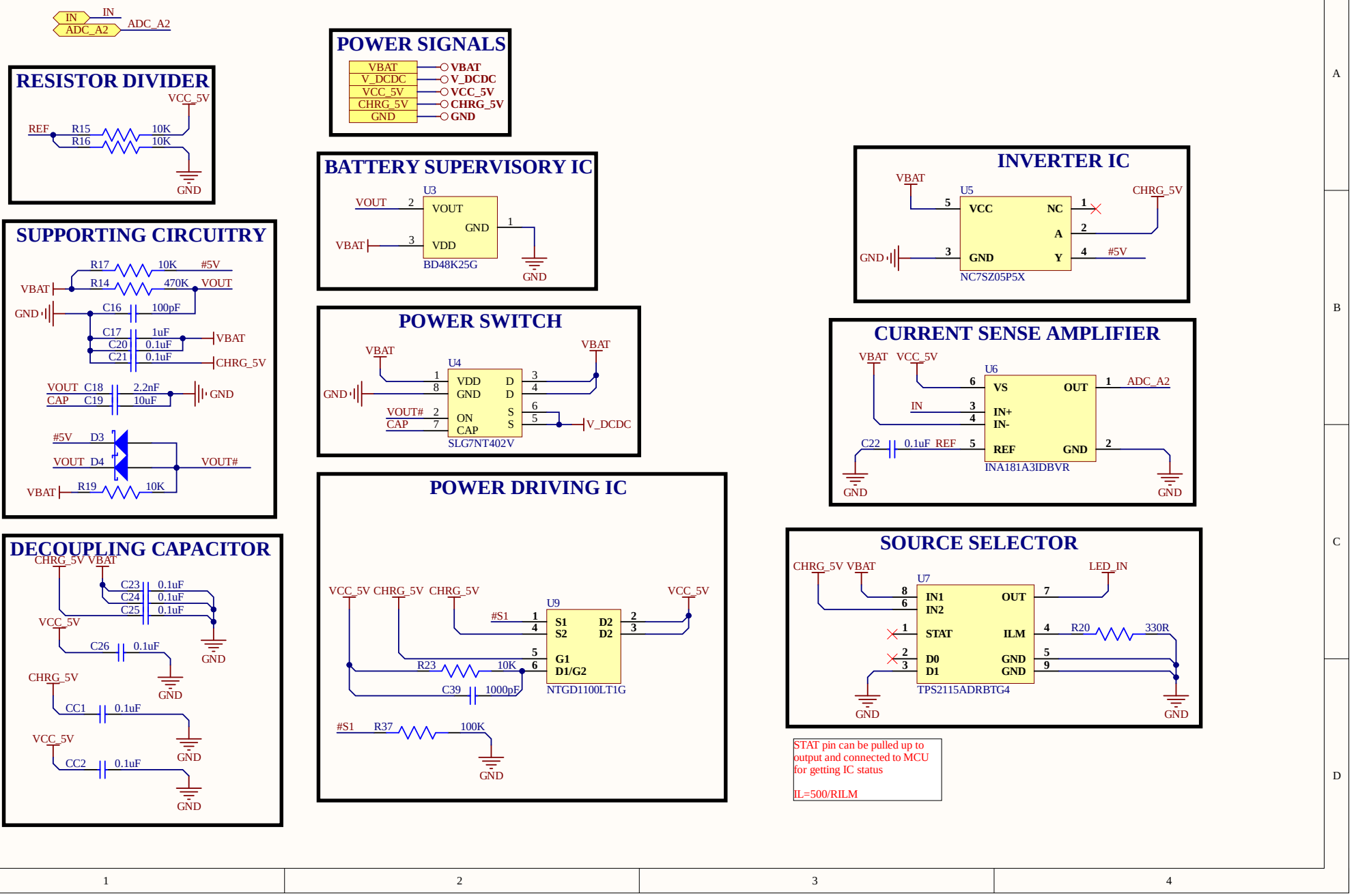
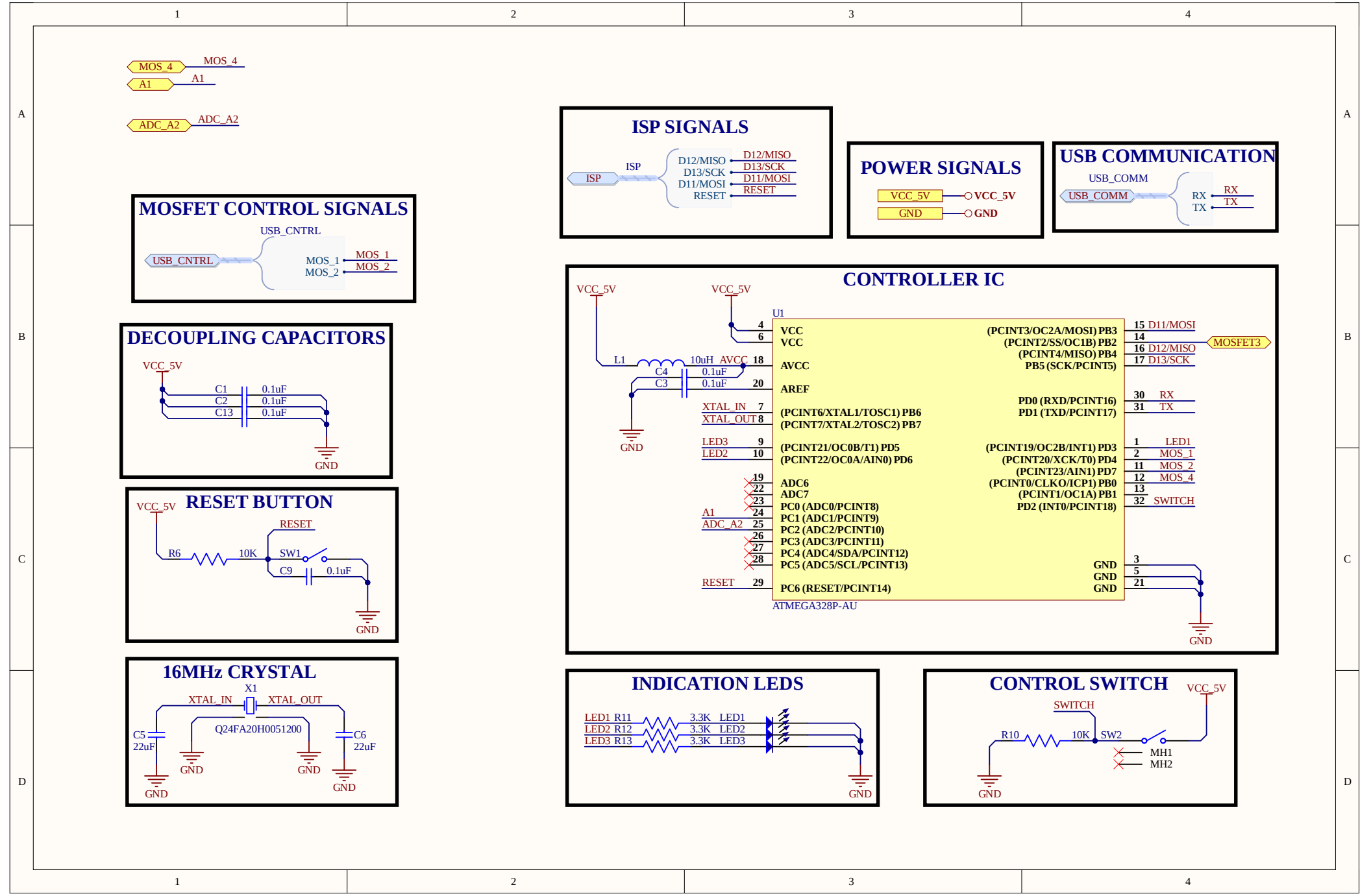
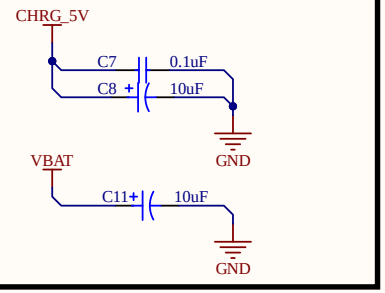

BATTERY CHARGING
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
CHRG_5V
VBAT
MOSFET3
GND
CHRG_5V
VBAT
GND
POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VBAT
CHRG_5V
V_DCDC
VCC_5V
ADC_A2
IN
GND
VBAT
CHRG_5V
V_DCDC
VCC_5V
IN
GND
DC REGULATOR
U_DCDC_Conv_Section
DCDC_Conv_Section.SchDoc
V_DCDC
VCC_5V
GND
V_DCDC
VCC_5V
GND
CONTROLLER
U_Controller_Section
Controller_Section.SchDoc
VCC_5V
MOSFET3
ADC_A2
MOS_4
A1
ISP
USB_CNTRL
USB_COMM
GND
VCC_5V
MOSFET3
ADC_A2
A1
ISP
USB_CNTRL
USB_COMM
GND
LAMP BLOCK
U_Lamp_Section
Lamp_Section.SchDoc
LED_IN
MOS_4
GND
LED_IN
MOS_4
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_5V
VBAT
CHRG_5V
IN
A1
ISP
GND
VCC_5V
VBAT
CHRG_5V
IN
A1
ISP
GND
USB BLOCK
U_USB_Section
USB_Section.SchDoc
VCC_5V
USB_CNTRL
USB_COMM
GND
VCC_5V
USB_CNTRL
USB_COMM
GND

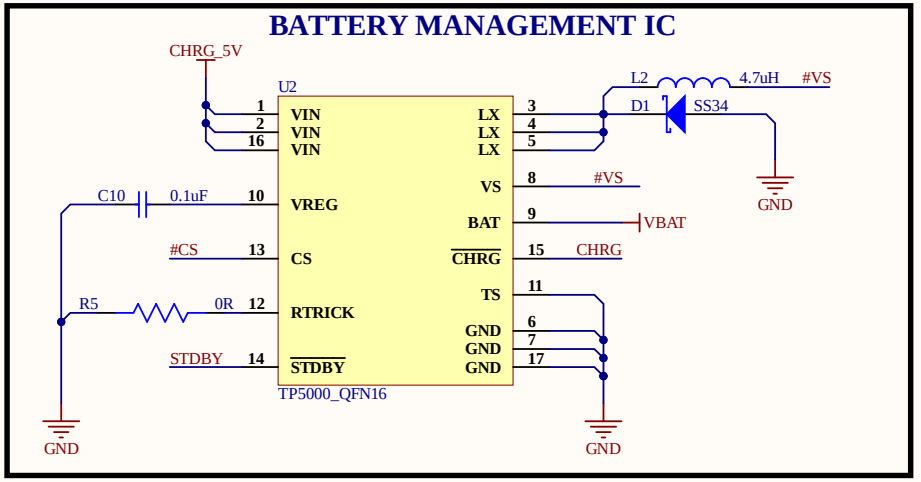

IN
IN
ADC_A2
ADC_A2
RESISTOR DIVIDER
VCC_5V
REF
R15
10K
R16
10K
GND
POWER SIGNALS
VBAT
V_DCDC
VCC_5V
CHRG_5V
GND
VBAT
V_DCDC
VCC_5V
CHRG_5V
GND
BATTERY SUPERVISORY IC
U3
VOUT
2
VOUT
GND
1
VBAT
3
VDD
BD48K25G
GND
INVERTER IC
VBAT
U5
CHRG_5V
5
VCC
NC
1
A
2
GND
3
GND
Y
4
#5V
NC7SZ05P5X
SUPPORTING CIRCUITRY
R17
10K
#5V
VBAT
R14
470K
VOUT
GND
C16
100pF
C17
1uF
C20
0.1uF
VBAT
C21
0.1uF
CHRG_5V
VOUT
C18
2.2nF
CAP
C19
10uF
GND
#5V
D3
VOUT
D4
VOUT#
VBAT
R19
10K
POWER SWITCH
VBAT
VBAT
U4
1
VDD
D
3
GND
8
GND
D
4
GND
VOUT#
2
ON
S
6
CAP
7
CAP
S
5
V_DCDC
SLG7NT402V
CURRENT SENSE AMPLIFIER
VBAT
VCC_5V
U6
6
VS
OUT
1
ADC_A2
IN
3
IN+
4
IN-
C22
0.1uF
REF
5
REF
GND
2
INA181A3IDBVR
GND
GND
DECOUPLING CAPACITOR
CHRG_5V
VBAT
C23
0.1uF
C24
0.1uF
C25
0.1uF
VCC_5V
GND
C26
0.1uF
GND
CHRG_5V
CC1
0.1uF
GND
VCC_5V
CC2
0.1uF
GND
POWER DRIVING IC
VCC_5V
CHRG_5V
CHRG_5V
U9
VCC_5V
#S1
1
S1
D2
2
4
S2
D2
3
5
G1
R23
10K
6
D1/G2
C39
1000pF
NTGD1100LT1G
#S1
R37
100K
GND
SOURCE SELECTOR
CHRG_5V
VBAT
U7
LED_IN
8
IN1
OUT
7
6
IN2
1
STAT
ILM
4
R20
330R
2
D0
GND
5
3
D1
GND
9
TPS2115ADRBTG4
GND
GND
STAT pin can be pulled up to output and connected to MCU for getting IC status
IL=500/RILM


MOS_4
MOS_4
A1
A1
ADC_A2
ADC_A2
MOSFET CONTROL SIGNALS
USB_CNTRL
USB_CNTRL
MOS_1
MOS_2
MOS_1
MOS_2
DECOUPLING CAPACITORS
VCC_5V
C1
C2
C13
0.1uF
0.1uF
0.1uF
GND
RESET BUTTON
VCC_5V
RESET
R6
10K
SW1
C9
0.1uF
GND
16MHz CRYSTAL
X1
XTAL_IN
XTAL_OUT
Q24FA20H0051200
C5
22uF
C6
22uF
GND
GND
GND
GND
ISP SIGNALS
ISP
ISP
D12/MISO
D13/SCK
D11/MOSI
RESET
D12/MISO
D13/SCK
D11/MOSI
RESET
POWER SIGNALS
VCC_5V
GND
VCC_5V
GND
USB COMMUNICATION
USB_COMM
USB_COMM
RX
TX
RX
TX
CONTROLLER IC
VCC_5V
VCC_5V
U1
VCC
VCC
AVCC
AREF
(PCINT6/XTAL1/TOSC1) PB6
(PCINT7/XTAL2/TOSC2) PB7
(PCINT21/OC0B/T1) PD5
(PCINT22/OC0A/AIN0) PD6
ADC6
ADC7
PC0 (ADC0/PCINT8)
PC1 (ADC1/PCINT9)
PC2 (ADC2/PCINT10)
PC3 (ADC3/PCINT11)
PC4 (ADC4/SDA/PCINT12)
PC5 (ADC5/SCL/PCINT13)
PC6 (RESET/PCINT14)
(PCINT3/OC2A/MOSI) PB3
(PCINT2/SS/OC1B) PB2
(PCINT4/MISO) PB4
PB5 (SCK/PCINT5)
PD0 (RXD/PCINT16)
PD1 (TXD/PCINT17)
(PCINT19/OC2B/INT1) PD3
(PCINT20/XCK/T0) PD4
(PCINT23/AIN1) PD7
(PCINT0/CLKO/ICP1) PB0
(PCINT1/OC1A) PB1
PD2 (INT0/PCINT18)
GND
GND
GND
ATMEGA328P-AU
L1
10uH
AVCC
C4
0.1uF
C3
0.1uF
GND
XTAL_IN
XTAL_OUT
LED3
LED2
A1
ADC_A2
RESET
MOSFET3
GND
INDICATION LEDS
LED1 R11
LED2 R12
LED3 R13
3.3K
3.3K
3.3K
LED1
LED2
LED3
GND
CONTROL SWITCH
VCC_5V
SWITCH
R10
10K
SW2
MH1
MH2
GND

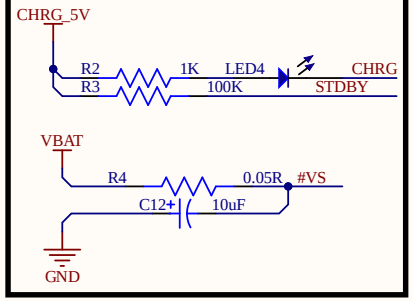
DECOUPLING CAPACITORS



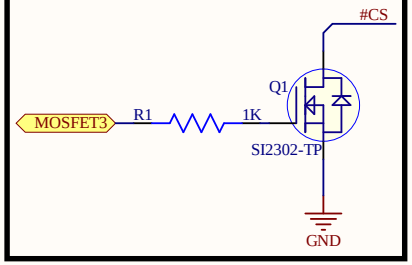
BATTERY MANAGEMENT IC



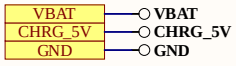
SUPPORTING CIRCUITRY



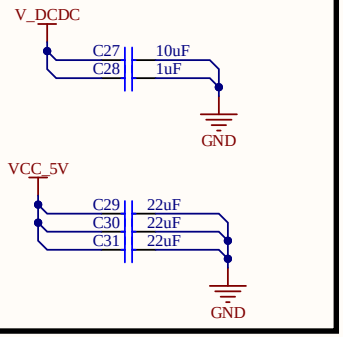
CONTROL CIRCUIT



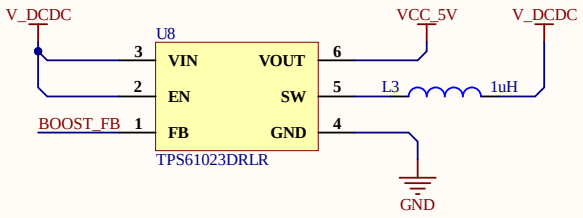
POWER SIGNALS



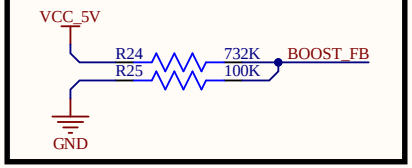
DECOUPLING CAPACITOR



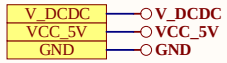
5V BOOST REGULATOR

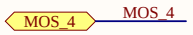


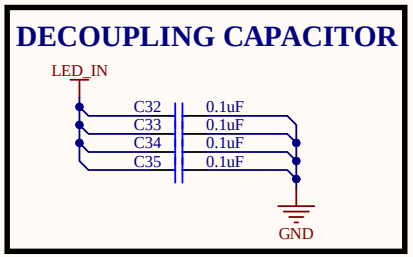
RESISTOR DIVIDER

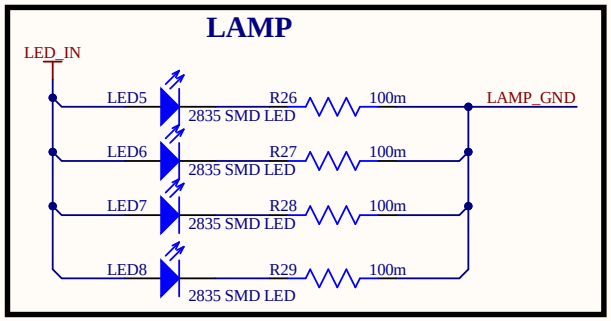


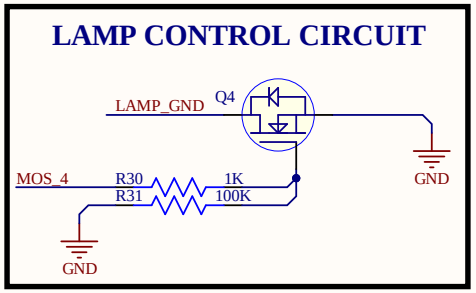
POWER SIGNALS

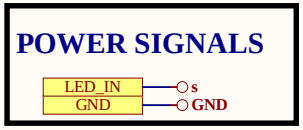



MOS_4
MOS_4


DECOUPLING CAPACITOR
LED_IN
C32
C33
C34
C35
0.1uF
0.1uF
0.1uF
0.1uF
GND


LAMP
LED_IN
LED5
LED6
LED7
LED8
R26
R27
R28
R29
100m
100m
100m
100m
2835 SMD LED
2835 SMD LED
2835 SMD LED
2835 SMD LED
LAMP_GND


LAMP CONTROL CIRCUIT
LAMP_GND
Q4
MOS_4
R30
R31
1K
100K
GND
GND


POWER SIGNALS
LED_IN
GND
s
GND

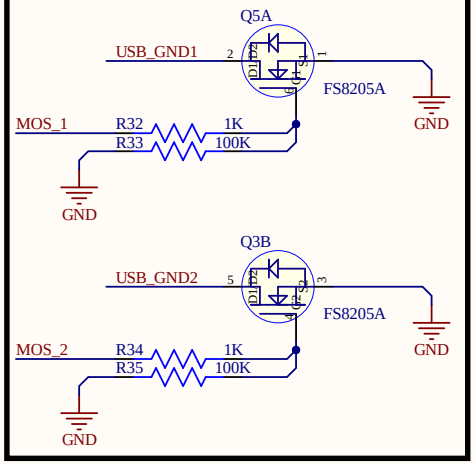
USB COMMUNICATION



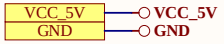
MOSFET CONTROL SIGNALS



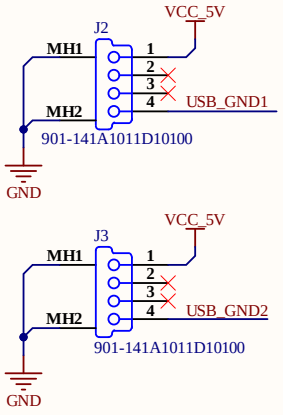
MOSFET CONTROL CIRCUIT



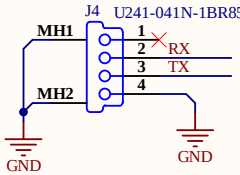
POWER SIGNALS

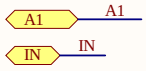


CHARGER OUTPUT



PROGRAMMING USB

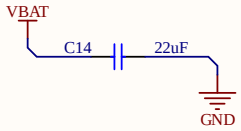




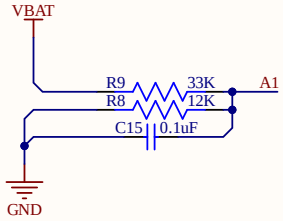
ISP SIGNALS



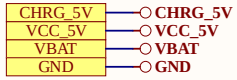
DECOUPLING CAPACITOR



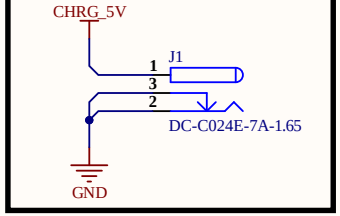
FEEDBACK NETWORK



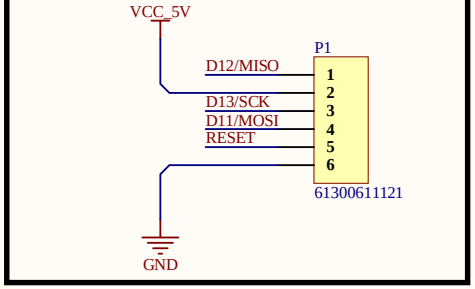
POWER SIGNALS



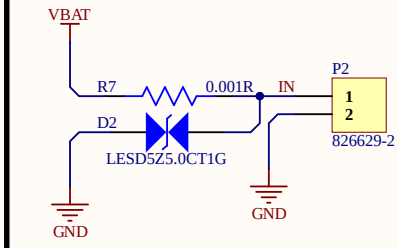
DC POWER HEADER



ISP PROGRAMMING HEADER



BATTERY HEADER



IN signal comes from Current sense amplifier circuitry